

GT Group: Building successful apps to improve customer experience with developer- friendly cloud resources

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[About GT Group](#)



About GT Group

Founded in 2012, GT Group specializes in mobile innovation for enterprise customers and consumers through multiple companies such as app development consultancy GreenTomato and start-up accelerator GT Studio. GreenTomato is the creator of Hong Kong's first moviegoing app Hong Kong Movie and voice messenger app Talkbox. The group has developed more than 500 apps, and counting.

Industries: Technology

GT Group migrate
Google Cloud to i
infrastructure arc
allowing for an a
flexible, and scal
platform and eco

create and develop
house and client

Location: Hong Kong

Google Cloud results

- Reduces 20% of VM costs with sizing recommendations from Compute Engine
- Accelerates app development with preemptible VMs at 80% of instance cost
- Fast deployment of VMs to accommodate 4X data storage increase during COVID-19
- Saves cloud spend by 30% with autoscaling capabilities of App Engine while managing sudden traffic peaks for seamless customer service

Receives
maximum
time
day
zero

Firestore (<https://cloud.google.com/firestore/>),

Google Cloud (<https://cloud.google.com/>)

Compute Engine

App Engine (<https://cloud.google.com/appengine/>)

Firebase (<https://firebase.google.com/>)

Cloud Firestore

(<https://cloud.google.com/firestore/>)

Cloud IAM (<https://cloud.google.com/iam/>)

Cloud IAM (<https://cloud.google.com/iam/>)

Compute Engine

(<https://cloud.google.com/compute/>)

Cloud IAM

Firebase (<https://firebase.google.com/>)

(<https://cloud.google.com/dialogflow-enterprise/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

(<https://workspace.google.com/>), Gmail

Cloud Vision API (<https://cloud.google.com/vision/>)

Cloud Vision API

Contact Center AI

(<https://cloud.google.com/solutions/contact-center/>)

Google Drive

Dialogflow Enterprise Edition

(<https://cloud.google.com/dialogflow-enterprise/>)

Google Meet

Google Workspace

(<https://workspace.google.com/>)

Google Sheets

Gmail

(<https://workspace.google.com/products/gmail/>)

Docs

(<https://workspace.google.com/products/docs/>)

Google Drive

(<https://workspace.google.com/products/drive/>)

Contact us (/contact)

What's the secret to success? The

Google Meet

(<https://workspace.google.com/products/meet/>)

GreenTomato (<https://www.gt.app>) for

Sheets

(<https://workspace.google.com/products/sheets/>)

asked. There's even a television s

Entrepreneurial Age, about the startup journey of

this award-winning mobile solutions consultancy.

That's because GreenTomato has created more than

500 apps, including Hong Kong's first moviegoing

app Hong Kong Movie (<https://hkmovie6.com/>), a voice

messenger app Talkbox (<https://teamwork.app>), and

chatbot builder Sanuker (<https://sanuker.com/>), now a

member company of GT Group

(<https://www.gtgroupcorp.com>), lends its experience

and a team of experts to help enterprise businesses

create the next hot app.

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*—Stephen Wong, co-
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Behind each enterprise app is a desire for businesses to connect with customers, deliver better service, and share their brand story. According to Stephen Wong, co-founder and Chief Technology Officer at GT Group, some companies have good ideas but lack the tech skills to bring the apps to market. GT Group bridges that gap with a full suite of mobile development services, from consultation to product development and analysis, so businesses don't need to start from scratch.

“As mobile developers, we need an agile network for research and development, whether it is analyzing data or testing out new project ideas for clients,” says Stephen. “[Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>) gives us easy access to

cloud resources to support our innovation and is the most flexible and convenient platform to create VMs and other services.”

Hong Kong Movie and Talkbox use Google Cloud and Google Cloud products such as [App Engine](#) (/appengine), [Compute Engine](#) (/compute), and [Firebase](https://firebase.google.com/) (https://firebase.google.com/) to host in-house and client apps and support app development for new projects. [Google Workspace](https://workspace.google.com/) (https://workspace.google.com/) helps developers work effectively with clients and each other, throughout the app development process.

Delighting millions of moviegoers with a smooth app experience

Green Tomato's pioneer apps Hong Kong Movie, launched in 2009, and Talkbox, in 2011, have seen much success over the years.

With more than three million downloads, Hong Kong Movie attracts more than 100,000 movie fans each day with reviews and an up-to-date showtime schedule in Hong Kong cinemas. The team also developed TIXIS, a point-of-sale system for online and onsite ticketing and F&B sales for cinemas.

To reduce system downtime on its legacy server, Hong Kong Movie migrated eight servers, including its backend and database servers, to Google Cloud in 2014. The team uses Compute Engine to support its API server, blog, and CI/CD server, while App Engine runs the website. Hong Kong Movie delivered uninterrupted service on its website during peak

demand, such as the launch of Avengers 4 in 2019, when more than 1,000 users rushed to buy tickets at the same time.

The team saves 80% of the cost by using preemptible VMs on Compute Engine for testing environments instead of purchasing additional hardware. It moved its development platform to Firebase in 2016 for fast deployment of features, without managing the infrastructure. Combined with Firestore (/firestore) NoSQL database, Hong Kong Movie developed new features such as campaigns and movie voting that queries user data.

As app data and users increased, Google Cloud provides Hong Kong Movie a scalable storage and processing power to control cost and resources. Jason Poon, assistant technical manager at Green Tomato says, “We've reduced 20% of VM costs by taking Compute Engine recommendations to adjust our resources based on actual usage.” At the same time, autoscaling on App Engine automatically deploys resources if the Hong Kong Movie website encounters sudden traffic peaks and scales down accordingly during low traffic periods, to save cloud spend by 30%.

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is no longer a bottleneck. We can deploy VMs in minutes when CPU usage doubled and data storage increased four-fold during COVID-19 to support communication between businesses users who needed to work from home.”

—Anita Wong, Lead Engineer - Backend, Talkbox Limited

Driving business growth and accelerating chat app development

Designed initially as a chat app for consumers, Talkbox gained popularity over time for its enterprise communication solution Teamwork Messenger,

which employees would use to communicate about work-related issues. However, the on-premises data center was not scalable enough to handle the increase in traffic as app usage expanded. To provide capacity, Talkbox had to add hardware and software licenses to its on-premises data center.

For each client, the IT team would spend weeks creating VMs from physical servers. Besides the time-consuming onboarding process, the team spent an average of 10 days a year in the data center resolving downtime from hardware issues.

The move to Google Cloud in 2015 gave Talkbox a fully managed infrastructure to handle scale without investing in more hardware and administrative burden. It now uses Compute Engine to create VMs in minutes and reduce service downtime to less than an hour.

“We couldn’t add hardware fast enough in the past to prevent service outages from sudden traffic spikes,” says Anita Wong, Backend Lead Engineer at Talkbox Limited. “On Compute Engine, the creation of VMs is no longer a bottleneck. We can deploy VMs in minutes when CPU usage doubled and data storage increased fourfold during COVID-19 to support communication between business users who needed to work from home.”

Talkbox developers also use short term VMs to quickly test and deploy new features such as MAM for bank-grade security, without the tedious task of preparing servers for availability. These preemptible

VMs are excess Compute Engine capacity and therefore cost less than normal instances.

With any app, user adoption is the real sign of success. Some customers start with a small-scale project on Talkbox and invest more when they see demand. Using [Cloud Monitoring \(/monitoring\)](#), the team provides customers with a clear picture of resource utilization and uptime, convincing customers to expand their project.

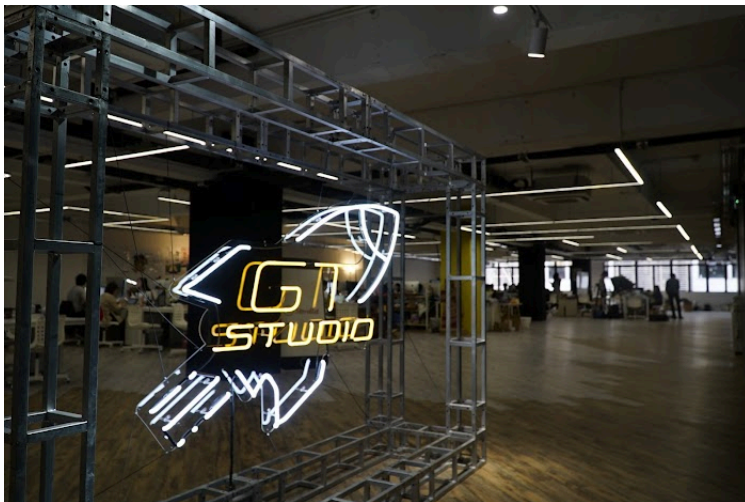
Protecting data and disaster recovery

Data security is top concern for many consumers and businesses alike. Hong Kong Movie protects customer data by using Google Cloud features such as [Cloud Identity and Access Management \(/iam\)](#) that prevents unauthorized access to Google Cloud resources. Every new staff at Hong Kong Movie is given a Google Workspace account that defines user permission on Google Cloud and the developer tools required to do their job. Once the employee resigns, their Google Workspace account, which is connected to all the resources and developer tools, is easily disabled within seconds.

Talkbox also uses the Google Workspace account to trace employee interaction on Google Cloud where necessary. For example, if cloud resources are accidentally deleted from Google Cloud, the administrator can check the logs to see who last accessed a document, as well as the full history of employees who read, wrote, or deleted any code. Using [Cloud Logging \(/logging\)](#), Talkbox can pinpoint

abnormal traffic in real time and remove any security loopholes before breaches occur.

To back up data, Talkbox uses Snapshots (/compute/docs/disks/create-snapshots), a Compute Engine feature, to create a copy of its VM each day. The legacy third-party software slows down app performance during backup. In case of unexpected issues such as power loss or human error, Talkbox can restore a VM from backup in minutes, so customers don't lose important information such as contacts and shared files.



Encouraging communication and collaboration for remote work

GreenTomato was an early adopter of Google Workspace, having onboarded in 2013. Today, the company's 200 employees in Hong Kong and Guangzhou use its productivity apps such as Gmail (<https://workspace.google.com/products/gmail/>) for emails, Google Drive (<https://workspace.google.com/products/drive/>) to share

resources, and Google Meet (<https://workspace.google.com/products/meet/>) for regular and ad hoc discussions.

Before Google Workspace, employees could only access emails from the file server in the office. They couldn't read and send emails or work files if they were out of the office for meetings or working from home. Keeping track of different document versions was also a problem. If more than one person worked on a shared file in the server simultaneously, teammates often overrode each other's changes or deleted information. With Docs or Google Sheets (<https://workspace.google.com/products/sheets/>), this is no longer an issue since everything is recorded in a timeline history.

Beyond collaboration, Google Workspace single sign-on enables employees to sign on with their Google accounts to authorized third-party apps, so they don't need to create and memorize multiple IDs and passwords. This means their work is not hampered by any accessibility issues to specific apps, and they are able to get their job done with all the tools easily accessible to them.

Engaging customers with conversational experiences

Sanuker is a chatbot consulting company in GT Group that uses Google Cloud to power its chatbot development. More than 2.8 million people subscribe to chatbots created by Sanuker.

To manage customer interaction for a Swedish luxury watch maker and its customers via a social media messaging app, Sanuker integrates Dialogflow (/dialogflow), part of Contact Center AI (/solutions/contact-center), with its in-house chatbot builder, Stella, for customer interaction. The chatbot called Junior Jones serves as a virtual assistant, engaging fans with personalized content. Dialogflow uses natural language processing (NLP) as its interactive interface to recognize user intent based on what the user types and share relevant data in the conversation.

More than 30,000 watch lovers have interacted with the bot between April 2020 and October 2020. Sanuker plans to enrich the watch brand chatbot experience with speech-to-text capability and integrate with Cloud Vision (/vision) for allowing customers to find watches by uploading images.

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AI and microservices on the horizon

As a whole, GT Group is expanding its Google Cloud capabilities to help employees gain more value from the platform.

The Talkbox team for example, plans to redesign its application as microservices running on Google Kubernetes Engine (/kubernetes-engine). Instead of deploying VMs one by one, Talkbox wants to automate the deployment process by creating containers from microservices with a single command line.

“We’re putting engineers through Google Cloud training and certifications so they can provide support to new members in their cloud journey,” says

Stephen. "With more than 500 apps in the group, we continue to look to Google Cloud to improve our service offering with new technologies such as big data and machine learning to better serve customers around the world."